

Exam 3

CH368 – Biophysics

Spring 2026

This exam is open book and open notes. You are allowed to use any resources available to you. However, you should not discuss the exam with anyone. The work you submit must be entirely your own.

This is provided as a MS Word file but please turn in your exam responses as a PDF on Canvas. Feel free to copy/paste the questions onto a new file or if you prefer to write your answers by hand, scan the paper and convert the images to a PDF.

Show all your work and write all your explanations in full sentences.

Honor Code: The core values of The University of Texas at Austin are learning, discovery, freedom, leadership, individual opportunity, and responsibility. Each member of the university is expected to uphold these values through integrity, honesty, trust, fairness, and respect toward peers and community.

SCORE: _____ / 150

1. **(20 points) Crowding and protein-protein interactions.** Consider the “hard sphere” model of crowding. As we discussed in class, the effect of crowding is to stabilize the more compact folded states of the protein. We also discussed protein dimerization in the context of partially folded and unfolded proteins. Based on this model, what do you predict the effect of crowding on protein dimerization to be? Hint: explain how you may expect the volume of a dimer to change compared to the two monomers.

Crowding favors dimerization. Two monomers occupy more total excluded volume than a dimer, because forming a dimer buries surface area and creates a more compact structure. This reduces excluded volume, increasing entropy of the crowders.

2. **(30 points) Force fields.** In molecular dynamics simulations, researchers typically refer to the “force field” in describing the parameters used to carry out a simulation.

- a. Explain in your own words what the force field represents.

A force field is a set of mathematical functions and parameters that approximate the potential energy of a system (bonded + nonbonded interactions), allowing forces and motions to be computed.

- b. Do you expect to be able to model chemical reactions with a traditional force field?

No. Traditional force fields cannot model chemical reactions because they assume fixed bonding and do not allow bond breaking/forming.

- c. How would you modify a force field to be able to study chemical reactions?

Use a reactive or quantum-based approach that allows bond orders and electronic structure to change.

- d. Induced dipole effects can be sometimes important to explain intermolecular interactions, but force fields do not capture these effects. Explain why traditional force fields fail to capture induced dipole interactions.

Traditional force fields use fixed point charges, so they cannot adjust electron distribution in response to the environment, and thus cannot capture induced dipoles (polarization effects).

2. **(20 points) Markov State Models.** In class we discussed Markov State Models in contrast with more traditional folding models such as the “funnel model” or the “chemical reaction model”. What are **two** advantages of the MSM over these other models (describe briefly)?

MSMs capture multiple folding pathways and intermediate states instead of assuming a single pathway or smooth funnel. They also allow estimation of long-timescale kinetics by combining many short simulations, making slow transitions accessible.

4. (20 points) Folding Coordinates. The folding landscape of a protein can be projected onto a single folding coordinate. Selecting an appropriate folding coordinate is important since the free energy landscape can be different depending on the coordinate selected. In class we described two models involving two different coordinate systems.

1. Explain in your own words what these two coordinates are and what the main difference is between the two.

The two coordinates are fraction of native contacts (Q) and RMSD. Q measures how many correct native interactions are formed, while RMSD measures overall structural deviation. The main difference is that Q tracks correct topology, whereas RMSD is purely geometric and can miss partially correct folding.

2. Propose a new coordinate system that could be used to potentially track folding of proteins and construct free energy surfaces from simulations.

You could track the burial of hydrophobic residues to capture protein compaction during folding.

5. (20 pts) Helical propensity. In class we showed a list of residues along with their propensities to form helices. How would you go about constructing such a table? Describe experiments/analyses that could be performed to determine the helical propensities of different amino acids.

You could construct this table by making a series of short peptides that are identical except for the residue being tested, then measuring how much helix each peptide forms. Helical content can be measured with circular dichroism (CD). By comparing the helix stability or fraction helix for peptides containing different amino acids at the same position, you can assign relative helical propensities.

6. (20 points) Fluid Mosaic Model. Describe in your own words the main characteristics of the fluid mosaic model of the membrane and outline its shortcomings. Specifically, what are the key aspects of membranes that are not described by the fluid mosaic model?

The fluid mosaic model describes the membrane as a 2D fluid lipid bilayer with proteins embedded and able to diffuse laterally, forming a dynamic “mosaic” of lipids and proteins held together by noncovalent interactions.

Its shortcomings are that it assumes the membrane is relatively homogeneous and freely diffusing, which is not fully accurate. It does not account for membrane heterogeneity, restricted protein mobility due to crowding effects, or specific lipid–protein interactions that organize membranes into structured domains.

7. (20 points) Membrane protein structure. In the protein databank (PDB) there are nearly 200 thousand structures of proteins. Out of these less than 2000 are membrane proteins, despite 30% of the genome coding for membrane proteins. How do we know that 30% of the genome code for membrane proteins if we have not yet determined their structures?

We know this from sequence-based predictions, not structures. By analyzing genomic sequences, we can identify features like hydrophobic transmembrane helices (via hydropathy plots) and signal peptides, which are characteristic of membrane proteins.